

House Price Prediction Using Machine Learning

Project Summary

This project aimed to develop a predictive model for real-estate prices utilizing machine-learning technology. The first stage consisted of performing exploratory analysis as well as cleaning the input dataset. Specifically, I searched for and removed extraneous, duplicate and/or missing records from the data that would impact the validity of the model; fortunately no issues were found and I did not have an inordinate number of items to validate. Since the data encompasses numerical and categorical variable types, I used One-Hot Encoding to convert the categorical variables into numerical values prior to using them in any machine learning model.

After the cleaning process was complete, I separated the data set into 80/20 splits (80% training and 20% test) for use in developing two predictive models; Linear Regression and Random Forest Regressor. To evaluate the accuracy of the predictions made by each of these models, I measured absolute (mean absolute error - MAE), root mean square (root mean square error - RMSE) and R^2 scores (a measure of how well the regression line fits the actual data points). The linear regression model yielded an R^2 score of approximately 0.65, while the random forest model yielded an R^2 of 0.61. Therefore, the linear regression model accounted for approximately 65% of the variation in price between homes, with this model being the most accurate of the two.

To present the results, I created a number of graphs. The histogram illustrates that there were more homes priced below and between 200k than there were homes that were priced above the 200k price point. The correlation matrix provided an indication of how the different variables correlate to each other and which show a strong positive or negative correlation. For instance, property tax has a strong negative correlation to both housing type (single-family residence versus multi-family residence), and housing price.